

Nov 12

Find

$$\sum_{n=0}^{\infty} (-2)^n \frac{\pi^{2n}}{6^{2n} (2n)!}$$

This is a power series where we've plugged in a value of x

$$\frac{\pi^{2n}}{6^{2n} (2n)!} = \frac{\pi^{2n}}{6^{2n}} \cdot \frac{1}{(2n)!} = \left(\frac{\pi}{6}\right)^{2n} \cdot \frac{1}{(2n)!} = \frac{\left(\frac{\pi}{6}\right)^{2n}}{(2n)!}$$

$$\hookrightarrow \sum_{n=0}^{\infty} (-2)^n \frac{\left(\frac{\pi}{6}\right)^{2n}}{(2n)!}$$

So this is $\cos(x)$ where $x = \frac{\pi}{6}$, so it just = $\cos\left(\frac{\pi}{6}\right)$

Recall that... $\cos(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

only even terms

ROC: $(-\infty, \infty)$ Alternating sign